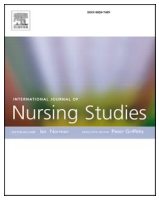
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The use of large language models in clinical documentation: A scoping review

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ABSTRACT

Background: Clinical documentation is essential for safe, high-quality care but has become increasingly complex, contributing to clinician burnout. Large language models offer potential to ease documentation by generating summaries, structuring data, and ensuring compliance. However, concerns remain regarding accuracy, bias, privacy, and regulatory risks.
Objective: To map current literature on large language models applications in clinical documentation, evaluating their benefits, limitations, and ethical considerations.
Information sources: Five electronic databases (i.e., PubMed, Scopus, CINAHL, Cochrane Library, and IEEE Xplore) covering peer-reviewed literature published in English between January 2009 and August 2025.
Methods: This scoping review followed Arksey and OMalleys framework and was reported in accordance with PRISMA-ScR guidelines. Screening, data extraction, and quality appraisal were conducted independently by multiple reviewers using Joanna Briggs Institute tools. Findings were synthesized using descriptive and narrative approaches.
Results: Forty-one studies met inclusion criteria, most originating from the United States. Large language models were primarily applied to clinical note generation, discharge summaries, and provider-patient encounter documentation. Key evaluation metrics included content accuracy, linguistic quality, and summarization performance. Large language models demonstrated potential to improve documentation efficiency and readability, with some studies reporting up to 40 % time savings. However, concerns about factual inaccuracies, hallucinations, and reduced performance in complex cases were common. Clinician perceptions were mixed. Some found notes generated by large language models helpful and well-structured, while others raised concerns about reliability, liability, and loss of clinical nuance. Ethical challenges included data privacy, security, and algorithmic bias, with varying levels of compliance across settings.
Conclusions: Large language models hold significant promise for enhancing clinical documentation by improving efficiency, standardization, and clarity. However, their safe and effective use requires rigorous attention to accuracy, ethical safeguards, and clinician trust. Integration must support, rather than supplant, clinical reasoning and patient-centered care. Co-design with clinicians, real-world evaluation, and artificial intelligence literacy are essential to ensure that these technologies augment, not erode, professional judgment and care quality.
Registration: Open Science Framework Registries (https://osf.io/m4h3q).

What is already known

- Clinical documentation is time-consuming and contributes to clinician burnout.
- Large language models may improve documentation but concerns over accuracy and ethics remain.
- Evidence on large language model use in clinical notes across settings is currently fragmented.

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What this paper adds

- Large language models can reduce clinician workload and improve documentation readability, but accuracy concerns persist.
- Open-source large language models offer greater transparency, while proprietary models show higher performance.
- Over-reliance on large language models for clinical documentation risks undermining clinical reasoning, highlighting the need for AI literacy.

1. Background

Clinical documentation is a core component of nursing and medical practice, ensuring continuity of care, supporting clinical decision-making, and serving as a legal and regulatory record of patient interactions. Accurate and comprehensive documentation is essential for effective communication among multidisciplinary healthcare teams, facilitating safe patient care transitions, and improving overall care quality. However, the increasing complexity of healthcare documentation has placed a substantial burden on clinicians, contributing to administrative overload and clinician burnout. For instance, clinicians managing patients with multiple comorbidities must document detailed assessments, medication reconciliations, and interdisciplinary care plans within electronic health records, requiring the synthesis of large volumes of information across diverse clinical domains within strict time constraints. Additionally, clinicians are tasked with documentation that supports regulatory and quality reporting, such as recording infection control measures, falls risk assessments, and patient education activities, which must meet detailed compliance standards and are often subject to external audit. Together, these complex documentation tasks highlight the growing administrative demands placed on healthcare providers (Gesner et al., 2022; Moy et al., 2021).

To address this challenge, artificial intelligence has been increasingly explored as a means to optimize clinical documentation processes. One of the most promising advancements in this field is the development of large language models which are a subset of artificial intelligence designed to process, understand, and generate human-like text. These models, including OpenAIs GPT-4, Googles BERT, Metas LLaMA, and healthcare-specific models like Med-PaLM, leverage transformer architectures, which are deep learning machine learning models designed to process sequences of text, to generate contextually relevant and coherent text (Nerella et al., 2024).

The application of large-scale large language models in clinical records can be traced back to the early 2010s, when recurrent neural networks and early deep learning models were first used to extract structured data from unstructured electronic health records (Shickel et al., 2018). A key milestone was the introduction of transformer-based models, beginning with Googles BERT in 2018, which enabled more accurate clinical concept extraction and contextual understanding of medical text (Devlin et al., 2019). This was followed by the development of domain-specific adaptations such as BioBERT (Lee et al., 2020) and ClinicalBERT (Alsentzer et al., 2019), which significantly advanced automated clinical note classification, cohort identification, and phenotyping. More recently, generative models such as GPT-3 (Brown et al., 2020) and its successors marked a paradigm shift by enabling not only information retrieval but also fluent text generation for tasks such as summarizing clinical encounters, drafting discharge summaries, supporting patient-facing documentation, and ensuring compliance with standardized documentation frameworks (Bedi et al., 2025; Omar et al., 2024). Collectively, these landmark achievements illustrate the rapid evolution from natural language processing systems focused on information extraction to large language models capable of producing coherent, human-like clinical documentation.

Despite these potential benefits, large language model-generated clinical documentation is still controversial regarding accuracy, reliability, and ethical implications, including factual inconsistencies, hallucination or incorrect information that can result in clinical errors, and training data biases that could worsen healthcare documentation quality disparities (Sallam, 2023). Ethical concerns related to patient privacy, data security, and regulatory compliance are significant, particularly given the sensitive nature of health information processed by large language models. Additionally, the lack of standardized evaluation frameworks for assessing large language model-generated clinical notes further complicates their integration into clinical practice (Meskó and Topol, 2023).

Given the increasing attention and rapid advancements in this area, there is a clear need for a comprehensive synthesis of existing evidence specifically targeting large language model use in nursing and medical documentation across diverse clinical contexts. While prior reviews have examined artificial intelligences broader role in healthcare and in nursing (Bedi et al., 2025; Hobensack et al., 2024; Yim et al., 2024), these reviews have largely treated large language models as a subset of broader artificial intelligence and have not provided an in-depth analysis of their specific role in clinical documentation. To address this knowledge gap, our scoping review aims to systematically map the empirical evidence on large language model applications in nursing and medical documentation, characterizing study designs, clinical contexts, and reported outcomes, while also identifying methodological challenges, ethical considerations, and directions for future research. Ultimately, this review seeks to inform clinical practice, policy development, and future research directions in clinical documentation leveraging large language models.

2. Methods

This scoping review was conducted following the methodological framework proposed by Arksey and O'Malley (2005) which has five stages: (1) Identify the research question; (2) Identify relevant studies; (3) Study selection; (4) Chart the data; and (5) Collate, summarize and report the results. The review's protocol was registered a priori at the Open Science Framework Registries (https://osf.io/m4h3q). The review was reported according to the Preferred Reporting Items for Systematic reviews and Meta-Analyses extension for Scoping Reviews checklist (Tricco et al., 2018) (Supplementary File 1).

2.1. Stage 1: identify the research question

This scoping review was conducted to explore the application and potential of large language models in optimizing clinical documentation across various clinical settings. Clinical documentation refers to the systematic process by which healthcare professionals record, update, and communicate patient-related information within the clinical record. For the purpose of this review, clinical documentation encompasses all forms of written or electronic entries made by nurses and doctors during or following patient care encounters, with the aim of supporting clinical decision-making, ensuring continuity of care, meeting legal and regulatory requirements, and facilitating interdisciplinary communication. The guiding research questions were:

(1) What types of large language model applications have been explored for use in doctors' and nurses' notes within in-patient settings?
(2) What are the main methodological approaches (e.g., qualitative, quantitative, mixed methods) used to study large language models in this context?
(3) What are the primary outcomes measured (e.g., accuracy, time-saving, clinician satisfaction)?
(4) What potential benefits of using large language models in nursing and medical documentation have been identified in the literature?

(5) What challenges or limitations are associated with the use of large language models for medical recordkeeping, according to the literature?

2.2. Stage 2: identify relevant studies

A comprehensive search of peer-reviewed literature was conducted across five electronic databases: PubMed, Scopus, CINAHL, Cochrane Library, and IEEE Xplore. The initial search was carried out in October 2024. To ensure the robustness of the search, relevant Medical Subject Headings, thesaurus terms and free-text keywords corresponding to two major concepts: (1) large language models and (2) clinical documentation (Table 1), were applied in each database. The search was restricted to articles published in English between January 2009 and August 2025 to ensure relevance to contemporary technological advancements.

An updated search was conducted on August 4, 2025 across all databases. Citation chaining was performed using BibliZap to identify additional studies citing or cited by the included articles. The search was further supplemented by manual screening of reference lists of eligible studies. A Bioinformatics Librarian from the University reviewed and validated the search strategy to ensure comprehensiveness and accuracy. The full search strategy is detailed in Supplementary File 2.

2.3. Stage 3: study selection

This review included papers that were:

(1) Original research published in peer-reviewed journals; and
(2) Studies that explore the application or potential of large language models in optimizing medical or nursing documentation, including clinical notes, consultation notes, shift summaries, and discharge/admission summaries in all clinical settings (outpatient, inpatient, emergency department, intensive care unit, adult, and pediatric care).

Table 1
Concept and keywords.

Concept 1 (LLM[a])/Keywords, MeSH Terms[b], MM[c]	Concept 2 (Clinical Documentation)/ Keywords, MeSH[b], MM[c]
Large Language Models (LLM)	Medical Recordkeeping
Natural Language Processing (NLP)	Healthcare Documentation
Natural Language Generator (NLG)	Medical Notes
Machine learning	Electronic Health Records (EHR)
Generative Pre-trained Transformer (GPT)	Doctors Notes
Google Bard	Nurses Notes
Transformer Model	Clinical Documentation
Generative Language Model	Healthcare Informatics
Deep Learning Language Model	Electronic Medical Records (EMR)
Pre-trained Language Model	Health Information Management
Neural Language Model	Medical Information Management
Neural Networks	Patient Information Management
Data to text	Health Records Management
ChatGPT	Medical Records Management
OpenAI	Patient Records Management
Computational linguistics	Patient Data Management
MM Natural Language Processing	Medical Documentation
	Interdisciplinary Notes
	Multidisciplinary Notes
	Shift summaries
	Nursing entries
	Medical entries
	Medical summaries
	electronic health records[MeSH Terms]
	MM Electronic Health Records
	MM Nursing Records

[a] LLM: Large Language Model.
[b] MeSH Terms: Medical Subject Headings.
[c] MM: Major Subject Headings.

This review excluded:

(1) Non-English articles;
(2) Studies focusing on artificial intelligence broadly, without specific reference to large language models;
(3) Studies utilizing large language models exclusively for radiology reports, laboratory investigations, pathology, medication administration records, or assessment checklists:
(4) Articles that have poor methodological or reporting quality; and
(5) Gray literature sources (preprints, dissertations, technical reports, or conference abstracts).

The search results from all databases were exported to Endnote 21 (The Endnote Team, 2013) for duplicate removal, then imported into Rayyan (Ouzzani et al., 2016) for the screening. The title and abstract screening was conducted independently by three authors (BW, HC and SBY). Articles meeting the eligibility criteria were retrieved for full-text review, which was conducted by three reviewers (BW, HC, and SBY). Discrepancies at any stage were resolved through team discussions among BW, HC, SBY and JS to ensure consensus.

2.3.1. Critical appraisal

To ensure the quality and rigor of the included studies, a critical appraisal was conducted using standardized Joanna Briggs Institute tools appropriate for each study design (i.e., observational studies and randomized controlled trials).

The selected articles were independently assessed by two pairs of reviewers (BW and HC; JS and SBY), with each pair appraising 50 % of the studies. Methodological quality, risk of bias, and overall reliability were evaluated using a structured appraisal tool based on established criteria (Bown and Sutton, 2010). For each criterion, reviewers assigned one of three possible ratings: Yes (Y) if the criterion was clearly met, No (N) if it was clearly not met, or Unclear (U) if there was insufficient information to make a judgment. The overall quality score for each study was calculated as the proportion of criteria rated "Yes" out of the total number of criteria assessed. Based on these scores, studies were categorized as follows:

(1) Low methodological quality (less than 40 %) – Excluded from this review;
(2) Medium methodological quality (40 to 70 %);
(3) High methodological quality (more than 70 %) (Bown and Sutton, 2010).

2.4. Stage 4: chart the data

A structured data extraction tool was collaboratively designed by the research team to define key variables essential for addressing the research questions (Levac et al., 2010). Before applying the tool to all selected studies, it was pilot-tested by BW and HC to assess its relevance and effectiveness. A finalized data extraction tool was then used to systematically collect the following study characteristics:

(1) Author, year, country of study;
(2) Study design and methodology;
(3) Clinical setting (outpatient, inpatient, emergency, intensive care, adult, pediatric);
(4) Type of medical documentation examined (e.g., clinical notes, discharge summaries, shift summaries);
(5) Large language models used and their specific applications;
(6) Outcomes related to accuracy, efficiency, ethical considerations, and impact on clinical workflows; and
(7) Identified challenges and limitations.

Data extraction was conducted independently by BW and JS, and cross-checked by HC and SBY to ensure accuracy and consistency.

2.5. Stage 5: collate, summarize and report the results

Descriptive numerical summary analysis was undertaken to characterize the included studies. The data was categorized to align with the research questions and extraction form (Arksey and O'Malley, 2005). The selected articles were reviewed multiple times to identify recurring patterns and key insights. To generate the narrative synthesis, the research team conducted iterative reading of the included studies and compared extracted data across studies to identify commonalities, divergences, and emergent themes. This process was guided by Braun and Clarkes (2006) thematic analysis approach, enabling us to organize findings into coherent categories that reflected patterns across study contexts, methodologies, and reported outcomes.

Consensus was achieved through structured team discussions. The preliminary summaries drafted by the first reviewer (BW) were reviewed by the other team members (HC, SBY and JS), who critically examined interpretations and raised alternative perspectives where appropriate. Disagreements were resolved through deliberation until agreement was reached, ensuring balanced interpretation and minimizing individual bias (Levac et al., 2010). The full research team then collectively refined and validated the final synthesized findings. All analyses were presented in a narrative format, supplemented with descriptive tables to enhance clarity and transparency.

3. Results

The PRISMA flow diagram summarizes the selection process (Fig. 1). Following the full-text review, 41 papers were assessed for methodological quality. The majority (n = 34) were rated as high quality (83 %), with the exception of seven studies, which were rated as medium quality (Supplementary File 3). Given their sufficient methodological rigor, all 41 papers were included in the synthesis, and a summary of the papers are presented in Supplementary File 4.

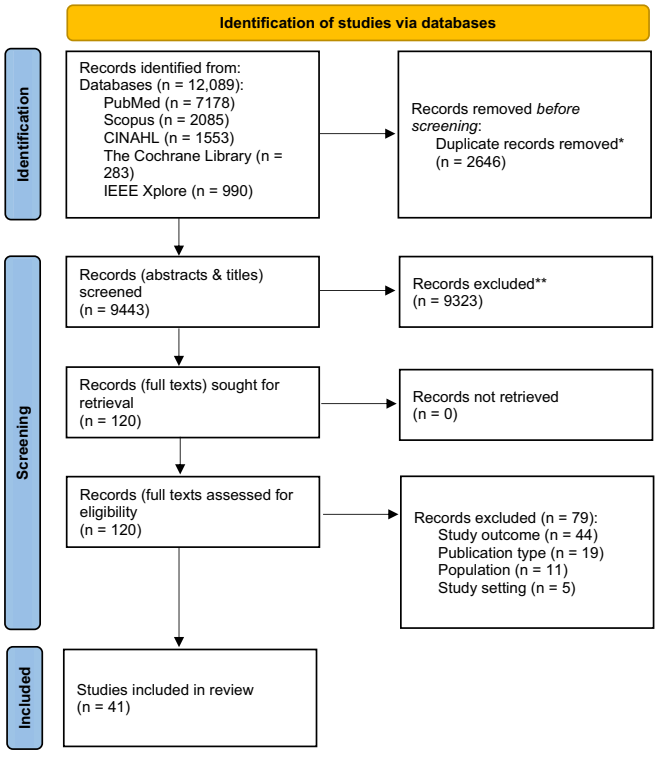


Fig. 1. PRISMA diagram.
*Rayyan was used to identify duplicates, crosschecked by a human.
**No automation tools used, excluded by humans.

3.1. Study characteristics

The characteristics of the included papers are described and cited in Table 2. Majority of the papers originated from the United States (US) (n = 17), followed by Europe (n = 7) and Asia Pacific (n = 7), with additional contributions from the United Kingdom (UK) (n = 5), and the Middle East (n = 3). Among the 41 studies, two were multi-country studies.

There was a predominance of observational studies (n = 37), three studies used a combination of qualitative and quantitative methodologies (Aali et al., 2025; Barak-Corren et al., 2024; Ganzinger et al., 2025), and only one study employed a randomized controlled trial design (Baker et al., 2024). Thirty studies utilized real-world data, defined as clinical documentation generated in actual care delivery settings (e.g., electronic health record notes, discharge summaries, consultation records), reflecting authentic clinical practice. In contrast, nine studies relied on simulated data, such as clinical vignettes, fabricated patient cases, or synthetic notes, which are commonly employed to test emerging technologies under controlled conditions before real-world implementation. This distinction is consistent with definitions in the broader informatics literature, where real-world data refers to health information routinely collected in clinical settings, while simulated data encompasses constructed or artificial datasets designed for evaluation and training purposes (Makady et al., 2017; Corrigan-Curay et al., 2018).

In studies (n = 39) which involved participants drafting or evaluating clinical notes, the participants comprised medical students (n = 25), residents (n = 39), physicians [n = 1020 (heavily inflated by Tailor et al.s 851 responses)], and nurses (n = 50). Their sample sizes ranged from two to 489 participants.

Table 2
Characteristics of included papers (n = 41).

Characteristics	n (%)
Study design	
Observational	37 (90)
Mixed-methods	3 (7)
Randomized controlled trial	1 (2)
Geographic location	
United States of America (USA)	17 (41)
Europe	7 (17)
Asia Pacific	7 (17)
United Kingdom (UK)	5 (12)
Middle East	3 (7)
Multi-country	2 (5)
Setting of primary studies	
Real-world data	30 (73)
Simulated cases	11 (27)
Large language model used/discussed[a]	
OpenAI Models [ChatGPT (unspecified)/ChatGPT-4/GPT-4/GPT-3.5/GPT-3.5-turbo-16k]	34
Google Models [BERT (and variants like UTH-BERT, RoBERYa); T5, FLAN-T5, T5-Booksum (long-text summarization); FLAN-UL2; PaLM2; PEGASUS-PubMed; Gemini 1.0]	9
Meta Models [LLaMA; LLaMA-2; LLaMA-2-Chat; Llama-2 7B; LLaMA 3 8b; LLaMA Derivatives (Alpaca, Vicuna, Med-Alpaca)]	4
Other Models [BART (Meta/Facebook AI); BART-Large-CNN; BLOOM-CLP-German; BART-LSG-conv; BART-DnC (divide-and-conquer fine-tuned); Claude; Mixtral 8x7B]	9

[a] As some studies use more than one large language model, the total count is over 41.

3.2. Large language models and their applications

Table 3 provides an overview of the large language models examined in the included studies, describing their types, general characteristics, and uses in medical recordkeeping. Both proprietary and open-source models were represented. Proprietary models included ChatGPT (OpenAI, 2022, 2023), Claude (Anthropic, 2023), and Gemini 1.0 Pro (Google, 2023). These models were primarily applied to tasks such as drafting patient histories (Aali et al., 2025; Barak-Corren et al., 2024; Gün and Aktaş, 2025; Indrakanti et al., 2025; Lu et al., 2025; Shemtob et al., 2025), supervisory notes (Barak-Corren et al., 2024), handoff briefs (Barak-Corren et al., 2024), discharge summaries (Falis et al., 2024; Ganzinger et al., 2025; Gün and Aktaş, 2025; Janota and Janota, 2025; Kharko et al., 2024; Kim et al., 2024; Li et al., 2025; Lim et al., 2025; Schwieger et al., 2024; Tung et al., 2024; Urquhart et al., 2024; Williams et al., 2025; Zaretsky et al., 2024), and summarizing provider-patient encounters.

Open-source models encompassed encoder- and decoder-based architectures as well as instruction-tuned variants. Encoder-based models such as BERT (Devlin et al., 2019), RoBERTa (Liu et al., 2019), and UTH-BERT (Ando et al., 2022) were commonly used for hospital course summarization (Hartman et al., 2023; Searle et al., 2023), classification, and structuring of nursing documentation (Moen et al., 2020). Decoder-based models such as LLaMA and LLaMA-2 (Meta AI, 2023a, 2023b), LLaMA-2-Chat (Meta AI, 2023b), and LLaMA-3 (Meta AI, 2024) were explored for discharge summaries (Heilmeyer et al., 2024; Urquhart et al., 2024) and specialist consultation notes (Heilmeyer et al., 2024). Instruction-tuned derivatives such as Vicuna (Vicuna Team, 2023), Alpaca (Taori et al., 2023), and Med-Alpaca (Han et al., 2023) were used in dialogue summarization (van Buchem et al., 2024; van Veen et al., 2024), progress notes, radiology reporting, and patient question answering (van Veen et al., 2024).

Hybrid encoder–decoder architectures, including BART (Lewis et al., 2019), T5 and its instruction-tuned variants FLAN-T5 and FLAN-UL2 (Chung et al., 2022; Raffel et al., 2020), and BLOOM-CLP-German (BigScience Workshop, n.d.), were applied to hospital course summaries (Hartman et al., 2023; Searle et al., 2023), clinical conversations (Navarro et al., 2025; Sezgin et al., 2024), and multilingual clinical documentation (Heilmeyer et al., 2024). More recent mixture-of-experts designs such as Mixtral 8x7B (Mistral AI, 2023) were also evaluated, particularly in discharge letter generation (Tailor et al., 2025).

3.3. Key evaluation metrics of large language models

Table 4 summarizes the key metrics used to assess large language models performance in medical recordkeeping. The most commonly reported evaluation metrics were content accuracy and completeness, linguistic quality and readability, and summarization performance, with additional domains capturing clinician assessments, comparisons of human versus large language model performance, and formatting or structural analysis.

Content accuracy and completeness were assessed through measures such as the proportion of sourced information (Ando et al., 2022), classification accuracy (Hartman et al., 2023; Moen et al., 2020), factual correctness (Dubinski et al., 2024; Ganzinger et al., 2025; Schwieger et al., 2024; Williams et al., 2025), and agreement rates between large language model-generated and physician-written reports (Heilmeyer et al., 2024; Indrakanti et al., 2025). Studies also examined error types such as omissions, additions, and incorrect facts, with some including harmfulness scoring adapting frameworks such as the Agency for Healthcare Research and Quality (AHRQ) Common Format Harm Scale (Kernberg et al., 2024; Williams et al., 2025). Accuracy of standardized nursing terminology (Johnson et al., 2025), recall, precision, and F1 scores (Bhayana et al., 2025; Dashti and Dashti, 2024; Indrakanti et al., 2025), and Source Chunking Approach for Large-scale inconsistency Evaluation fidelity score fidelity scores for handoff notes (Hartman

Table 3
Summary of Large Language Models in included papers.

Large Language Model name	Type & architecture	Specialization	Use in medical recordkeeping
Propriety models ChatGPT[a] (unspecified), ChatGPT-4, GPT[b]-4, GPT[b]-3.5, GPT[b]-3.5-turbo-16k, GPT[b]-4 Turbo	Transformer-based models (OpenAI, GPT[b] architecture)	Conversational AI[c], text generation, summarization, general NLP[d]	• Drafting history of present illness (Baker et al., 2024; Bhayana et al., 2025) • Physician supervisory notes (Barak-Corren et al., 2024) • Hand-off briefs (Barak-Corren et al., 2024) • Patient summaries (Aali et al., 2025; Barak-Corren et al., 2024; Gün and Aktaş, 2025; Indrakanti et al., 2025; Lu et al., 2025; Shemtob et al., 2025) • Discharge letters and summaries (Falis et al., 2024; Ganzinger et al., 2025; Gün and Aktaş, 2025; Janota and Janota, 2025; Kharko et al., 2024; Kim et al., 2024; Li et al., 2025; Lim et al., 2025; Schwieger et al., 2024; Tung et al., 2024; Urquhart et al., 2024; Williams et al., 2025; Zaretsky et al., 2024) • Summary of provider-patient encounter (Ding et al., 2025; Misurac et al., 2025; Patel et al., 2025; van Buchem et al., 2024; van Veen et al., 2024) • Summary of progress notes (Gün and Aktaş, 2025; van Veen et al., 2024) • Summary of radiology reports (van Veen et al., 2024) • Summary of patient questions (van Veen et al., 2024) • Nursing care plan (Johnson et al., 2025; Ju et al., 2025) • Narrative nursing documentation (Ju et al., 2025)

(continued on next page)

Table 3 (*continued*)

Large Language Model name	Type & architecture	Specialization	Use in medical recordkeeping
			• Post-operation reports (Lim et al., 2025) • Patient-education materials (Gün and Aktaş, 2025; Mohan et al., 2024)
Claude	Decoder-only Transformer (Anthropic, 2023)	Conversational AI[c], text processing and summarization, content generation, general NLP[d]	• Post-operation reports (Lim et al., 2025) • Discharge summaries (Lim et al., 2025)
Gemini 1.0 Pro	Transformer-based decoder-only (Google, 2023)	Conversational AI[c], general NLP[d] tasks, coding support	• Summary of provider-patient encounter (Patel et al., 2025)
Open-source models			
BERT	Encoder-based Transformer (Google, 2018)	Text classification, NLP[d] processing	• Hospital course summaries (Hartman et al., 2023; Searle et al., 2023) • Nursing point-of-care documentation (Moen et al., 2020) • Spelling correction for Non-English notes (Dashti and Dashti, 2024)
Robustly Optimized BERT[e] Approach (RoBERTa)	Encoder-based Transformer (Google, 2018)	Excels in text classification and other supervised NLP[d] tasks after fine-tuning	• Inpatient handoff notes (Hartman et al., 2024)
UTH-BERT	Fine-tuned BERT[e] for medical text (Japan, UTH-Clinical BERT[e])	Medical text processing, clinical NLP[d]	• Discharge summaries (Ando et al., 2022)
LLaMA[f], LLaMA[f]-2	Decoder-based Transformer (Meta AI, 2023a, 2023b)	General text processing, open-source LLM[g], general NLP[d]	• Discharge summaries (Heilmeyer et al., 2024; Urquhart et al., 2024) • Non-English Specialist consultation notes (Heilmeyer et al., 2024) • Inpatient handoff notes (Hartman et al., 2024)
LLaMA[f]-2-Chat	Fine-tuned LLaMA[f]-2 for conversational AI[a] (Meta AI, 2023a, 2023b)	Conversational AI[c], optimized for chat interactions	• Non-English Specialist consultation notes (Heilmeyer et al., 2024)
LLaMA[f]-3	Decoder-based Transformer (Meta AI, 2024)	General text processing, open-source LLM[g], general NLP[d]	• Discharge summaries (Li et al., 2025)
Vicuna	Fine-tuned LLaMA[f] model for conversational AI[a]	Dialogue-focused LLM[g], user-centric conversational AI[c]	• Summary of provider-patient encounter (van Buchem et al., 2024; van Veen et al., 2024)
Alpaca	Fine-tuned LLaMA[f] (Stanford)	Instruction-tuned, open-source NLP[d]	• Summary of provider-patient encounter (van Buchem et al., 2024; van Veen et al., 2024) • Summary of progress notes (van Veen et al., 2024) • Summary of radiology reports (van Veen et al., 2024) • Summary of patient questions (van Veen et al., 2024)
Med-Alpaca	Fine-tuned Alpaca model for medical Q&A[h]	Medical Q&A[h], healthcare-specific NLP[d]	• Summary of provider-patient encounter (van Buchem et al., 2024; van Veen et al., 2024) • Summary of progress notes (van Veen et al., 2024) • Summary of radiology reports (van Veen et al., 2024) • Summary of patient questions (van Veen et al., 2024)
BART[i]	Hybrid Encoder-Decoder Model (Facebook AI, 2019)	Summarization, text refinement, document structuring	• Hospital course summaries (Hartman et al., 2023; Searle et al., 2023) • Summaries of clinical conversations (Navarro et al., 2025; Sezgin et al., 2024)
BLOOM[j]-CLP[k]-German	Transformer-based BLOOM[j] model, fine-tuned for German medical text	Medical text summarization for German clinical documentation	• Non-English Specialist consultation notes (Heilmeyer et al., 2024)
T5[l]	Text-to-Text Transfer Transformer (Google)	Text transformation, summarization, machine translation	• Summaries of clinical conversations (Navarro et al., 2025; Sezgin et al., 2024)
FLAN[m]-T5[l]	Fine-tuned T5[l] for instruction-based tasks (Google)	Instruction-based NLP[d], fine-tuned for diverse text tasks	• Summary of provider-patient encounter (van Buchem et al.,

(*continued on next page*)

Table 3 (*continued*)

Large Language Model name	Type & architecture	Specialization	Use in medical recordkeeping
			2024; van Veen et al., 2024) • Summary of progress notes (van Veen et al., 2024) • Summary of radiology reports (van Veen et al., 2024) • Summary of patient questions (van Veen et al., 2024)
FLAN[m]-UL2[n]	Advanced version of FLAN[m]-T5[l] with more parameters (Google)	Advanced multi-turn reasoning, instruction-based AI[c]	• Summary of provider-patient encounter (van Buchem et al., 2024; van Veen et al., 2024) • Summary of progress notes (van Veen et al., 2024) • Summary of radiology reports (van Veen et al., 2024) • Summary of patient questions (van Veen et al., 2024)
Mixtral 8x7B	Transformer-based Mixture of Experts (Mistral AI, 2023)	General-purpose LLM[g], reasoning tasks, coding, multilingual coverage	• Discharge letter and summaries (Tailor et al., 2025)

[a] ChatGPT: Chat Generative Pre-trained Transformer.
[b] GPT: Generative Pre-trained Transformer.
[c] AI: Artificial Intelligence.
[d] NLP: Natural Language Processing.
[e] BERT: Bidirectional Encoder Representations from Transformers.
[f] LLaMA: Large Language Model Meta AI.
[g] LLM: Large Langugage Model.
[h] Q&A: Question and Answer.
[i] BART: Bidirectional and Auto-Regressive Transformers.
[j] BLOOM: BigScience Large Open-science Multilingual.
[k] CLP: Cross-Lingual and Progressive transfer learning.
[l] T5: Text-to-Text Transfer Transformer.
[m] FLAN: Fine-tuned Language Net.
[n] UL2: Unifying Language Learning paradigms.

et al., 2024) were also reported.

Linguistic quality and readability were commonly evaluated using clarity, coherence, and grammatical correctness (Holstead, 2024; Janota and Janota, 2025). Readability scores included the Flesch–Kincaid Grade Level, Flesch–Kincaid Grade Level, Reading Ease, Coleman–Liau Index, and Simple Measure of Gobbledygook (Barak-Corren et al., 2024; Hartman et al., 2023; Urquhart et al., 2024; Zaretsky et al., 2024; Lim et al., 2025; Tailor et al., 2025). Patient comprehension was assessed using tools such as Patient Education Materials Assessment Tool (PEMAT) Understandability Scale and satisfaction with plain-language summaries (Holstead, 2024; Zaretsky et al., 2024; Tailor et al., 2025). Conciseness was measured via word counts and physician feedback (Hartman et al., 2023; Tung et al., 2024; Urquhart et al., 2024; van Veen et al., 2024; Williams et al., 2025), and sentiment or empathy analysis was applied to examine tone and patient-centeredness (Kharko et al., 2024).

Summarization performance was frequently assessed using Recall-

Table 4
Key evaluation metrics assessing large language model's performance in medical recordkeeping.

Category	Evaluation metrics	Details
Content accuracy and completeness	Proportion of sourced information (Ando et al., 2022)	Amount of data derived from inpatient records.
	Classification accuracy (Hartman et al., 2023; Moen et al., 2020)	Correct classification of diagnoses, treatments, and follow-ups.
	Factual correctness (Dubinski et al., 2024; Ganzinger et al., 2025; Schwieger et al., 2024; Williams et al., 2025)	Rated by physicians (e.g., % correctly reported elements; error counts per summary).
	Types of errors (Kernberg et al., 2024; Williams et al., 2025)	Omissions, additions, incorrect facts; harmfulness scoring [adapted Agency for Healthcare Research and Quality (AHRQ) Common Format Harm Scale].
	Agreement rates (Heilmeyer et al., 2024; Indrakanti et al., 2025)	Alignment between AI and physician-written reports [e.g., Intraclass Correlation Coefficient (ICC) reliability].
	Accuracy of standardized nursing terminology (Johnson et al., 2025)	Alignment with North American Nursing Diagnosis Association/ Nursing Interventions Classifications/Nursing Outcomes Classification (NANDA/NIC/NOC) labels.
	Recall, precision, F1 score (Bhayana et al., 2025; Dashti and Dashti, 2024; Indrakanti et al., 2025)	Extraction performance for clinical history parameters, spelling correction, neurovascular data.
	Source Chunking Approach for Large-scale inconsistency Evaluation (SCALE) fidelity score (Hartman et al., 2024)	Assesses faithfulness of Emergency Medicine (EM)-to-inpatient handoff notes.
Linguistic quality and readability	Clarity, coherence, and grammar (Holstead, 2024; Janota and Janota, 2025)	Assessed through manual review and rating scales.
	Readability scores (Barak-Corren et al., 2024; Hartman et al., 2023; Urquhart et al., 2024; Zaretsky et al., 2024; Lim et al., 2025; Tailor et al., 2025)	Flesch-Kincaid, Reading Ease, Coleman-Liau Index, Simplified Measure of Gobbledygook (SMOG).
	Patient comprehension (Holstead, 2024; Zaretsky et al., 2024; Tailor et al., 2025)	Patient Education Materials Assessment Tool (PEMAT), satisfaction with plain-language summaries.
	Conciseness (Hartman et al., 2023; Tung et al., 2024; Urquhart et al., 2024; van Veen et al., 2024; Williams et al., 2025)	Evaluated by word count and physician feedback.
	Sentiment and empathy analysis (Kharko et al., 2024)	Assesses tone, empathy, and patient-centeredness of generated notes.
Summarization metrics	Organization and sequencing (Urquhart et al., 2024; Shemtob et al., 2025)	Quality of structuring and logical order of events.
	Recall-Orientated Understudy for Gisting Evaluation (ROUGE) and Bilingual Evaluation Understudy (BLEU) Scores	Measures precision, recall, and F1-score; token-level metrics.

(*continued on next page*)

Table 4 (*continued*)

Category	Evaluation metrics	Details
	(Hartman et al., 2023; Searle et al., 2023; Sezgin et al., 2024; van Buchem et al., 2024; Li et al., 2025; Aali et al., 2025)	
	Bidirectional Encoder Representations from Transformers (BERT) Score (van Buchem et al., 2024; Ding et al., 2025; Li et al., 2025; Aali et al., 2025)	Semantic similarity with reference summaries.
	Unified Holistic Evaluation for Unified Multimodel Understanding and Generation (UniEval) metrics (Navarro et al., 2025)	Coherence, fluency, consistency, relevance, overall quality.
	Comparison of prompting methods (Kim et al., 2024; Aali et al., 2025)	Statistical tests [Analysis of Variance (ANOVA), Tukey's test] to assess effectiveness.
Physician and clinician assessments	Correlation with input length (Kim et al., 2024)	Relationship between input size and quality.
	Efficiency of note generation (Barak-Corren et al., 2024; Lu et al., 2025)	Time required and perceived effort
	Satisfaction with artificial intelligence summaries (Barak-Corren et al., 2024; Tailor et al., 2025)	Rated on scales of 0–10 or Likert-type survey.
	Level of acceptance (Clough et al., 2024)	Physicians' acceptance ('Yes'/'No').
	Comparison with expert summaries (van Veen et al., 2024; Shemtob et al., 2025)	Physician preference ratings and quality scores.
	Potential for medical harm (van Veen et al., 2024; Williams et al., 2025)	Review of risks posed by incorrect artificial intelligence-generated content; harmfulness scoring.
	Burnout and fulfillment outcomes (Misurac et al., 2025)	Stanford Professional Fulfillment Index scores.
Human versus large language model performance comparison	Time metrics (van Buchem et al., 2024; Lu et al., 2025; Lim et al., 2025)	Manual vs automated summarization/ documentation time.
	Physician documentation quality (PDQI-9) (van Buchem et al., 2024; Patel et al., 2025; Kernberg et al., 2024)	Scores on accuracy, completeness, organization.
	Hallucination occurrences (Urquhart et al., 2024; Schwieger et al., 2024; Williams et al., 2025)	Frequency of fabricated information; clinical significance.
	Comparative reviewer ratings (Baker et al., 2024; Schwieger et al., 2024)	Blinded expert scoring of AI vs clinician notes.
Formatting and structural analysis	Inter-rater reliability (Urquhart et al., 2024; Li et al., 2025)	Agreement among physicians reviewing AI output [e.g., Intraclass Correlation Coefficient (ICC)].
	Subject heading accuracy (Moen et al., 2020)	Correct assignment of headings.
	Grouping coherence (Moen et al., 2020)	Logical structuring of related sentences.
	Effectiveness of paragraph merging (Moen et al., 2020)	Impact on clarity and readability.
	Structure and format adherence (Schwieger et al., 2024; Indrakanti et al., 2025)	Scored against templates; structured versus free-text alignment.

Oriented Understudy for Gisting Evaluation (ROUGE) and BiLingual Evaluation Understudy (BLEU) scores (Hartman et al., 2023; Searle et al., 2023; Sezgin et al., 2024; van Buchem et al., 2024; Li et al., 2025; Aali et al., 2025). Bidirectional Encoder Representations from Transformers Score (BERTScore) was employed to assess semantic similarity (van Buchem et al., 2024; Ding et al., 2025; Li et al., 2025; Aali et al., 2025). Some studies applied Unified Evaluation Framework (UniEval) metrics for coherence, fluency, and relevance (Navarro et al., 2025) or compared prompting methods statistically using Analysis of Variance and Tukeys tests (Kim et al., 2024; Aali et al., 2025). Event organization and sequencing were also assessed (Urquhart et al., 2024; Shemtob et al., 2025).

Physician and clinician assessments included evaluations of efficiency and effort (Barak-Corren et al., 2024; Lu et al., 2025), satisfaction with large language model-generated summaries (Barak-Corren et al., 2024; Tailor et al., 2025), and levels of acceptance (Clough et al., 2024). Comparisons with expert-written summaries assessed quality and preferences (van Veen et al., 2024; Shemtob et al., 2025). Risks of medical harm were explicitly evaluated (van Veen et al., 2024; Williams et al., 2025), and one study examined clinician outcomes such as burnout and fulfillment using the Stanford Professional Fulfillment Index (Misurac et al., 2025).

Human versus large language model performance comparisons looked at time metrics for documentation (van Buchem et al., 2024; Lu et al., 2025; Lim et al., 2025), documentation quality scores using the Physician Documentation Quality Instrument (PDQI-9) (van Buchem et al., 2024; Patel et al., 2025; Kernberg et al., 2024), and hallucination occurrences (Urquhart et al., 2024; Schwieger et al., 2024; Williams et al., 2025). Blinded comparative ratings between large language model and clinician outputs were also conducted (Baker et al., 2024; Schwieger et al., 2024).

Formatting and structural analysis evaluated inter-rater reliability (Urquhart et al., 2024; Li et al., 2025), subject heading accuracy, grouping coherence, and paragraph merging effectiveness (Moen et al., 2020). Other measures included adherence to structural templates and format consistency (Schwieger et al., 2024; Indrakanti et al., 2025).

3.4. Impact of large language model use in clinical documentation

3.4.1. Efficiency and time savings

Several studies reported that large language models improved documentation efficiency by reducing the time required for summarization and note generation (Barak-Corren et al., 2024; Dubinski et al., 2024; Lu et al., 2025; van Buchem et al., 2024). For instance, ChatGPT decreased documentation time by 40 % for supervisory notes in complex cases (Barak-Corren et al., 2024), while large language model-generated summaries required significantly less editing time than manually written notes (van Buchem et al., 2024). In neurosurgery, ChatGPT markedly reduced the time to complete discharge summaries and operative reports compared to speech-recognition software (Dubinski et al., 2024). Similarly, GPT-4o generated hematology records in 40 s versus 15 min for residents, without significant loss of quality (Lu et al., 2025).

In nursing contexts, large language model-assisted documentation tools reduced documentation time by approximately 40 % compared with traditional methods (Ju et al., 2025), while large language model-generated care plans reduced the burden of structuring nursing documentation (Johnson et al., 2025; Moen et al., 2020). In psychiatry, ChatGPT generated discharge summaries in 15 min compared with 90 min for psychotherapists, with comparable coherence and grammar scores (Janota and Janota, 2025). GPT-4-based systems also reduced documentation time by approximately 14 min per discharge summary in palliative care settings (Gün and Aktaş, 2025).

Beyond time metrics, studies also reported reduced cognitive load and enhanced workflow efficiency. Ambient artificial intelligence systems such as Nabla generated two-thirds of notes in real-world ambulatory clinics, leading to reductions in burnout and improved work-life

balance (Misurac et al., 2025). Similarly, large language models produced structured summaries that accelerated clinician preparation for consultations (Indrakanti et al., 2025; Shemtob et al., 2025).

Many studies, however, were limited by small sample sizes (Barak-Corren et al., 2024; Clough et al., 2024; Urquhart et al., 2024) and reliance on simulated scenarios (Holstead, 2024; Lim et al., 2025; Tung et al., 2024), limiting generalizability. Other limitations included variability in large language model performance across specialties (Ding et al., 2025), inefficiency in capturing temporally complex information (Indrakanti et al., 2025), and rigidity or verbosity in large language model-generated notes requiring further refinement for clinical usability (van Buchem et al., 2024; Schwieger et al., 2024; Williams et al., 2025).

3.4.2. Accuracy and reliability

While large language models demonstrated efficiency gains, concerns about accuracy and reliability were prominent. Studies identified frequent errors, including factual inaccuracies, omissions, and hallucinations. For example, ChatGPT introduced erroneous information in 36 % of documents (Baker et al., 2024), and large language model-generated discharge summaries sourced only 61 % of their content from inpatient records, relying on physician memory for the remainder (Ando et al., 2022). Other studies quantified error types using harmfulness scoring (Williams et al., 2025) and identified clinically significant omissions and distortions (Kernberg et al., 2024). Hallucinations, where generated text appeared plausible but was incorrect, were reported across multiple domains, including fabricated diagnoses, treatment details, and laboratory results (Kim et al., 2024; Urquhart et al., 2024; Schwieger et al., 2024).

Performance varied by clinical complexity. High accuracy was reported in constrained tasks, such as spelling correction for non-English clinical text (Dashti and Dashti, 2024), classification of nursing notes (Moen et al., 2020), and standardized discharge summaries using structured templates (Heilmeyer et al., 2024; Indrakanti et al., 2025). However, accuracy declined in complex medical cases. For example, neurosurgical reports showed reduced factual accuracy compared to human-written text (Dubinski et al., 2024), hematology records generated by GPT-4o omitted key details (Lu et al., 2025), and patient-friendly discharge summaries in Korean included fabricated content requiring correction (Kim et al., 2024).

Several studies also highlighted variability in performance across clinical contexts. GPT-4o performed reliably for neurovascular consultations but struggled with temporally complex patient trajectories (Indrakanti et al., 2025). Psychiatric discharge summaries generated by ChatGPT achieved coherence and readability but lacked diagnostic nuance (Janota and Janota, 2025). Similarly, large language model-generated palliative care summaries captured basic events but missed subtle clinical reasoning (Gün and Aktaş, 2025).

3.4.3. Clinician workload and experience with large language model-generated documentation

Large language model-generated documentation was perceived as improving clarity, organization, and completeness, which may alleviate cognitive burden and burnout. For example, ambient artificial intelligence note systems decreased documentation burden and were associated with lower burnout and higher fulfillment scores on the Stanford Professional Fulfillment Index (Misurac et al., 2025). In addition, large language model-generated summaries were rated as acceptable by all general practitioners in one study (Clough et al., 2024), and were frequently described as easier to read and edit compared with manually written notes (van Buchem et al., 2024; Lim et al., 2025).

Clinicians also recognized potential benefits in supporting interdisciplinary workflows. Nurses reported that large language model-assisted nursing care plans and progress notes reduced time spent on repetitive tasks and improved alignment with standardized terminologies (Johnson et al., 2025; Ju et al., 2025; Moen et al., 2020). In psychiatry, psychotherapists noted that large language model-generated discharge summaries reduced workload but lacked the nuance and empathy expected in mental health records (Janota and Janota, 2025).

Despite these positive perceptions, barriers to adoption were consistently identified. Concerns regarding variability in accuracy across specialties and prompting strategies (Ding et al., 2025; Sezgin et al., 2024), limited ability to convey clinical reasoning (Barak-Corren et al., 2024; Gün and Aktaş, 2025), medico-legal liability (Schwieger et al., 2024), and resistance to reliance on automated tools (Urquhart et al., 2024). Variability in tone and empathy was especially concerning in patient-facing documentation (Kharko et al., 2024; Tailor et al., 2025), with some clinicians cautioning that such limitations may undermine patient trust.

3.4.4. Patient-centered benefits

Several studies highlighted the potential of large language model-generated documentation to enhance patient comprehension and engagement. Large language model-generated discharge letters and summaries consistently demonstrated higher readability scores than clinician-authored notes, with reductions in Flesch–Kincaid Grade Level and improved plain-language clarity (Kim et al., 2024; Zaretsky et al., 2024; Tailor et al., 2025). Patients rated these documents as easier to understand, and Patient Education Materials Assessment Tool Understandability Scores indicated higher levels of understandability (Holstead, 2024; Zaretsky et al., 2024; Tailor et al., 2025). In addition, large language model-supported systems facilitated patient education by generating tailored explanations of diagnoses, treatments, and follow-up instructions (Mohan et al., 2024; Gün and Aktaş, 2025).

Beyond readability, studies also demonstrated that large language models could support patient engagement through empathetic tone and accessibility. Kharko et al. (2024) showed that patient-facing notes generated by the large language model conveyed greater empathy, while Misurac et al. (2025) reported that ambient artificial intelligence tools freed up clinicians time for direct patient interaction.

Despite these benefits, concerns about completeness and safety persisted. Some studies found that large language model-generated summaries omitted key medical details, posing potential risks for patient safety (Urquhart et al., 2024; van Veen et al., 2024). Studies also cautioned that the trade-off between readability and comprehensiveness could compromise safety if oversimplification led to omission of critical information (Schwieger et al., 2024; Williams et al., 2025). Clinician oversight was consistently emphasized as essential to safeguard accuracy while retaining the accessibility advantages of large language model-generated documentation.

3.5. Ethical and practical challenges

Ethical concerns related to privacy, data security, and bias were widely reported. Several studies highlighted risks associated with non-Health Insurance Portability and Accountability Act compliant or cloud-based large language model tools, which could compromise patient confidentiality and data protection (Clough et al., 2024; Holstead, 2024; Sezgin et al., 2024). Other studies emphasized the need for robust data anonymization and encryption to protect patient confidentiality (Ando et al., 2022; Hartman et al., 2023). Urquhart et al. (2024) specifically noted the risk that ChatGPT-4 could retain patient data for future model training, potentially violating privacy regulations.

Bias in large language model-generated documentation was also a recurring issue. Outputs often reflected disparities embedded in training datasets, risking inequities in clinical communication and care delivery (Kharko et al., 2024; Tailor et al., 2025). For example, Williams et al. (2025) observed harmfulness scoring in erroneous summaries, which raised concerns that bias and misinformation could amplify clinical risks.

Liability issues were also highlighted, as clinicians remain responsible for errors in large language model-generated documentation, raising questions about medical-legal accountability (Barak-Corren

et al., 2024). Studies in psychiatry and emergency care noted that the lack of explainability in large language model-generated notes complicated responsibility for errors (Barak-Corren et al., 2024; Schwieger et al., 2024).

To mitigate ethical risks, some studies implemented safeguards such as the use of the Health Insurance Portability and Accountability Act-compliant platforms for large language model-generated patient summaries (Zaretsky et al., 2024) and secure institutional networks rather than cloud-based large language model services (Sezgin et al., 2024; Indrakanti et al., 2025). Compliance with General Data Protection Regulation was emphasized in European studies, where non-proprietary models were preferred to enhance data security (Heilmeyer et al., 2024). Other strategies included manual verification by multiple clinicians, use of harmfulness scoring tools, and structured templates to constrain free text generation (Kernberg et al., 2024; Williams et al., 2025).

4. Discussion

This scoping review highlights the growing interest in the application of large language models in clinical documentation, particularly for generating and summarizing doctors and nurses notes. The findings indicate that large language models offer potential benefits in improving documentation efficiency, reducing clinician workload, and enhancing patient comprehension. Studies in this review reported significant time savings and improved standardization in medical records, with large language model-generated summaries often achieving high readability scores (Barak-Corren et al., 2024; Zaretsky et al., 2024). However, accuracy remains a major concern, with frequent reports of factual errors, omissions, and hallucinations, particularly in complex clinical scenarios. While large language models demonstrated strong performance in structured tasks such as hospital course summaries and spelling correction, their reliability in real-world applications has room for improvement due to inconsistencies in generated outputs and the need for human oversight. Ethical concerns, including privacy risks, data security, and embedded biases, were widely discussed, emphasizing the importance of regulatory compliance and clinician validation.

4.1. Using open-source and proprietary large language models in clinical documentation

The adoption of open-source large language models presents distinct advantages, including increased transparency, adaptability, and community-driven improvements. Open-source models allow healthcare institutions to tailor the technology to specific clinical contexts and documentation standards without reliance on proprietary solutions. This adaptability enhances potential integration with existing electronic health record systems, enabling institutions to comply with privacy regulations more effectively, particularly in regions governed by stringent policies such as the European Unions General Data Protection Regulation which prohibits the transfer of protected health data out of the European Union territory (Heilmeyer et al., 2024).

However, several drawbacks must be considered. Open-source large language models often require significant technical expertise to develop, implement, maintain, and optimize, which may be resource-intensive, especially in smaller healthcare settings without robust IT support (Sezgin et al., 2024). Additionally, open-source models might lag behind proprietary models in terms of training data size, computational power, and sophistication, potentially compromising performance, accuracy, and reliability—critical factors in clinical documentation (Bedi et al., 2025). Regulatory compliance and accountability may also pose challenges, as responsibility for data breaches or errors in documentation could become complex without centralized vendor support.

Conversely, proprietary large language models provide robust, commercially-supported solutions with comprehensive technical support, which can significantly reduce the burden of implementation and maintenance (Naik et al., 2024). Proprietary models are often trained on extensive, well-curated datasets with higher computational resources, potentially ensuring higher accuracy, reliability, and overall performance, particularly beneficial in complex clinical documentation scenarios. Additionally, proprietary vendors often offer integrated solutions specifically designed for healthcare settings, facilitating smoother interoperability with existing electronic health record systems (Kirk et al., 2024).

Nevertheless, proprietary models pose significant limitations, including less transparency about their training processes and datasets, raising concerns about potential biases, fairness, and accountability. The lack of transparency can make it challenging for healthcare providers to understand, audit, and mitigate biases or inaccuracies (Dennstädt et al., 2025). Furthermore, reliance on proprietary solutions may increase dependency on specific vendors, leading to higher costs and reduced flexibility in system customization and scalability. Regulatory compliance might also become complex if vendors fail to maintain adherence to evolving data security and privacy standards, potentially exposing institutions to risks associated with breaches or misuse of patient information (Kirk et al., 2024).

4.2. Implementation of large language model technology in clinical settings

Effective integration of large language model technology into clinical documentation practices demands a modular and flexible approach. Modularity refers to breaking down complex technology implementations into smaller, manageable units or modules, allowing healthcare organizations to introduce and test each unit incrementally (Kushniruk et al., 2013). This approach enables targeted application of large language model solutions to specific documentation tasks, such as discharge summaries, patient handoffs, or shift documentation, without overwhelming staff or disrupting existing clinical routines. Implementing large language models in a modular fashion facilitates gradual clinician acceptance, as healthcare teams can gradually adapt to technology-enhanced processes, refining and troubleshooting them on a smaller scale before broader adoption (Kushniruk et al., 2013; Tung et al., 2024).

By employing a modular approach, healthcare providers can individually evaluate and continuously improve each documentation task (Reddy et al., 2021), such as initially applying large language model technology solely for discharge summaries. Staff can assess and refine performance in this focused area, identifying and resolving challenges early, which enhances robustness and reliability (van Buchem et al., 2024). Furthermore, modularity significantly simplifies integrating large language model technology within current electronic health record systems, as it does not necessitate major system overhauls or extensive retraining. This incremental method minimizes the disruption to clinicians' established workflows and supports smoother, more manageable transitions (Sezgin et al., 2024).

Importantly, the integration of large language models may also influence clinicians' documentation behavior, including their reliance on, interaction with, and trust in generated content. Early studies suggest that while large language models can reduce documentation time and cognitive load, they may also shift clinicians roles from active documentation creators to reviewers and editors, potentially altering their sense of ownership and vigilance over the clinical narrative (Urquhart et al., 2024; van Veen et al., 2024). Clinician acceptance of large language models hinges not only on perceived usefulness and ease of use but also on confidence in the systems accuracy and alignment with clinical reasoning. Therefore, implementation strategies must address not only technological integration but also the behavioral, cognitive, and cultural dimensions of clinical practice (Busch et al., 2025; Choudhury and Chaudhry, 2024).

As large language models become more integrated into clinical workflows, there is growing concern about their impact on clinical cognition and reasoning. Clinical documentation is not a passive task; it

is a generative act that reinforces critical thinking, reflection, and clinical decision-making. For novice practitioners especially, the act of writing notes helps consolidate assessment data, apply clinical frameworks, and develop diagnostic acumen (Richardson et al., 2022). Delegating this cognitive labor to large language models risks eroding these educational opportunities and may inadvertently deskill future clinicians. Over-reliance on large language models could lead to "automation complacency," where clinicians accept the large language models outputs without sufficient verification, potentially compromising patient safety (Parasuraman and Manzey, 2010). Moreover, the use of generic large language model-generated templates may promote a homogenized, one-size-fits-all approach to documentation, undermining the individualization of patient care (Ando et al., 2022). To counter this, health professional education must actively integrate artificial intelligence literacy while reinforcing documentation as a pedagogical and cognitive process. Clinicians should be equipped not only to use artificial intelligence tools, but also to critique and refine them, preserving the intellectual rigor that defines high-quality, person-centered care (Rony et al., 2024; Sun et al., 2023).

4.3. Future directions for research and practice

Future research should prioritize addressing gaps identified by nurses and other frontline clinicians by conducting rigorous real-world studies that evaluate the long-term effectiveness, usability, and ethical implications of open-source versus proprietary large language models, specifically within nursing and diverse healthcare contexts. Studies should investigate how modular versus comprehensive system-wide integration impacts nursing workflows, documentation efficiency, clinician satisfaction, and ultimately patient safety and outcomes.

In parallel, urgent attention must be paid to how large language models may influence clinical cognition and professional development. Clinical documentation is more than an administrative task. It is an intellectual process that underpins clinical reasoning, reflection, and decision-making. The act of writing, particularly for novice clinicians, is integral to developing the cognitive skills necessary for high-quality care. As large language models take on increasing responsibility for drafting and summarizing clinical notes, there is a risk that over-reliance may deskill clinicians, diminish engagement with patient narratives, and erode critical thinking over time. Future research should explore how large language model-generated documentation affects clinicians' cognitive work, including their ability to synthesize information, justify decisions, and construct personalized care plans.

To mitigate these risks, educational institutions and healthcare organizations must evolve curricula and training programs to incorporate artificial intelligence literacy while actively preserving opportunities for learners to engage in thoughtful, manual documentation. This includes emphasizing simulation-based training, critical appraisal of artificial intelligence outputs, and pedagogical strategies that highlight writing as a mode of clinical reasoning. Moreover, longitudinal studies are needed to examine how large language model adoption affects not only workflow outcomes, but also clinician cognition, confidence, and learning trajectories across career stages.

Equally critical is the co-design of large language model solutions with active involvement from nurses, educators, and other healthcare stakeholders throughout the model development process. Engaging end-users in the design, testing, and refinement of large language model tools ensures that the technology reflects real-world documentation practices, aligns with interdisciplinary workflows, and preserves the complexity of clinical narratives. Such participatory approaches enhance contextual relevance, safety, and trust—key drivers of successful adoption.

Finally, developing nursing-sensitive benchmarks for evaluating the accuracy, readability, completeness, and ethical implications of large language model-generated nursing documentation remains essential. Collaborative efforts among nurses, nurse educators, informaticians, technology developers, and regulatory bodies will be critical in establishing practical frameworks and guidelines. These should ensure that large language model technologies meaningfully improve documentation, support decision-making at the bedside, and maintain the highest standards of patient-centered, ethically sound care.

4.4. Limitations of review

While this review provides a comprehensive synthesis of studies on the use of large language models in clinical documentation, several limitations must be acknowledged. The predominance of cross-sectional and observational designs further limits insights into the long-term impact of large language models on clinical workflows, clinician workload, and patient safety.

The variability in evaluation metrics across studies, including different scoring systems for accuracy, readability, and efficiency, complicates direct comparisons and limits the ability to establish standardized benchmarks for large language model performance in clinical documentation. Additionally, most studies were conducted in controlled or simulated environments, raising concerns about real-world applicability, particularly regarding integration into electronic health record systems and clinician workflows.

Potential publication and selection biases must also be acknowledged, as studies with positive findings may be more likely to be published, while non-English studies were excluded, limiting insights into large language model performance across diverse healthcare contexts.

Finally, the rapid advancement of large language model technology means that findings may quickly become outdated as newer models with improved accuracy, bias reduction, and contextual understanding emerge. Future research should prioritize longitudinal studies, standardized evaluation methods, and real-world implementation to ensure ongoing relevance and applicability.

5. Conclusion

Large language models represent a promising advancement in clinical documentation, with the potential to improve efficiency, standardization, and comprehension across healthcare settings. However, this review highlights that realizing these benefits requires careful attention to accuracy, ethical safeguards, and clinician trust. Importantly, the integration of large language models must not compromise the cognitive processes and critical thinking that underlie high-quality, patient-centered care. Future efforts should prioritize co-design with clinicians, rigorous real-world evaluation, and educational strategies that preserve clinical reasoning while promoting artificial intelligence literacy. By balancing innovation with ethical and professional integrity, large language models can be harnessed to support rather than replace the human expertise at the heart of healthcare.

CRediT authorship contribution statement

Brigitte Fong Yeong Woo: Writing – original draft, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Kenrick Cato:** Writing – review & editing, Supervision. **Hannah Cho:** Writing – review & editing, Formal analysis. **Sang Bin You:** Writing – review & editing, Formal analysis. **Jiyoun Song:** Writing – review & editing, Supervision, Formal analysis.

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Declaration of competing interest

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Appendix A. Supplementary data

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